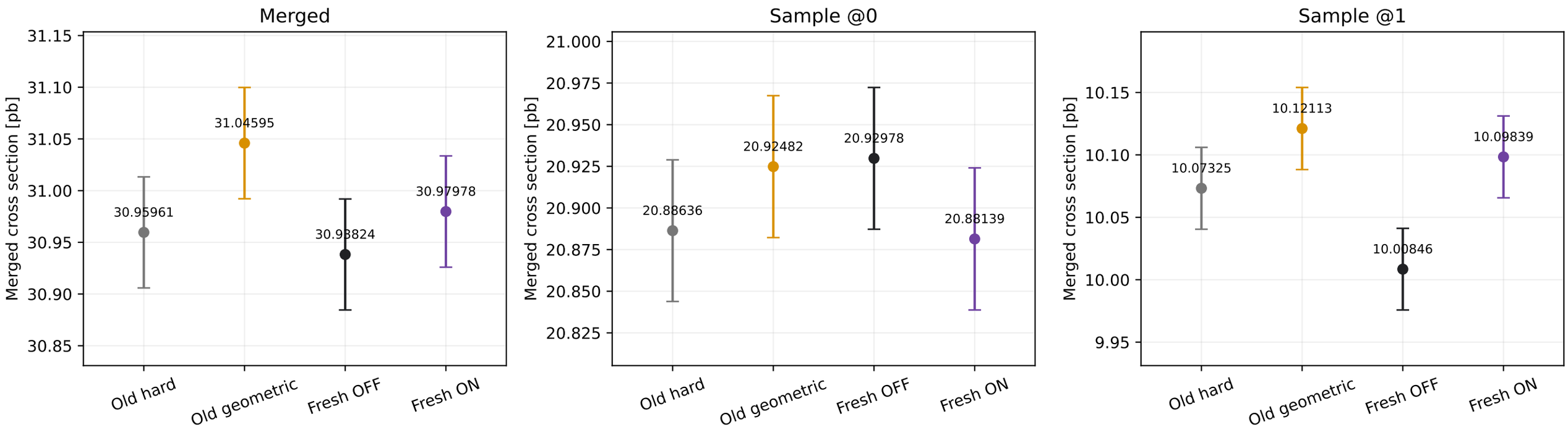
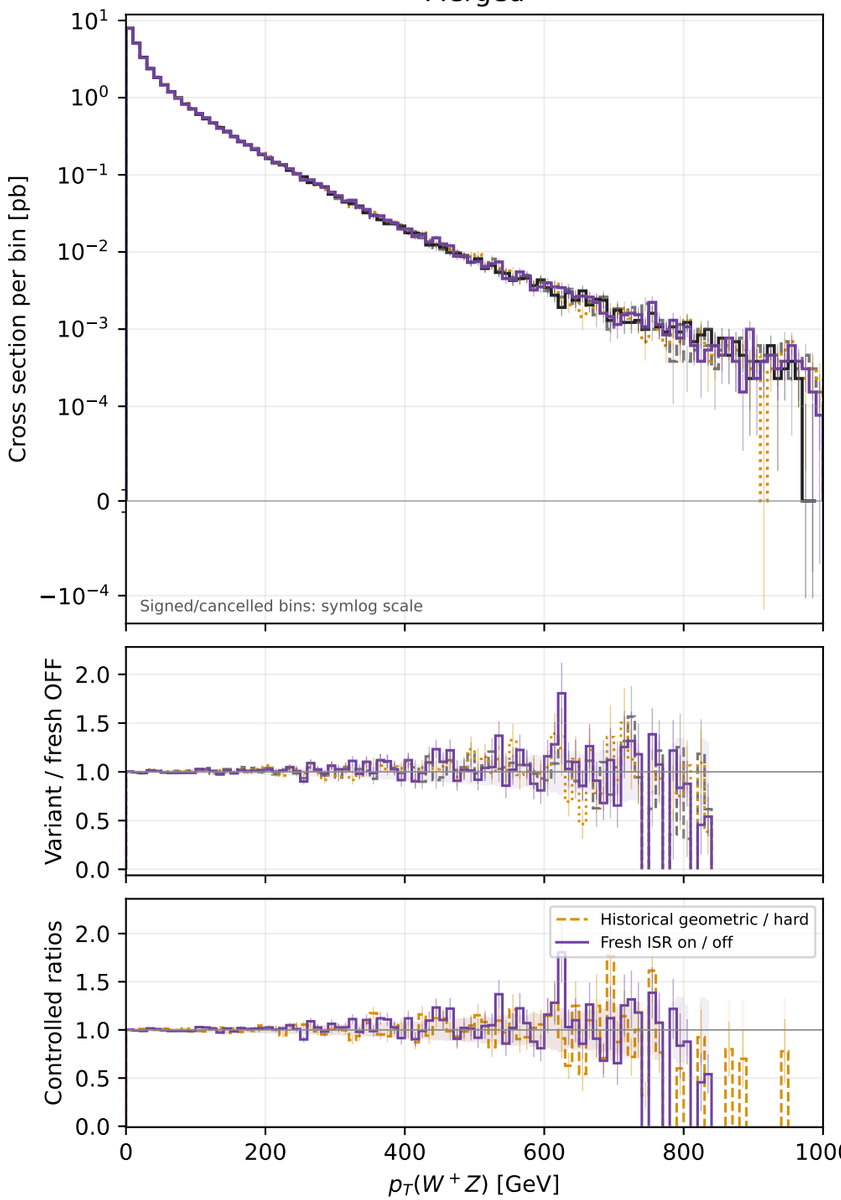

WZ + 0,1 jets: absolute rates for all four prescriptions
Merged
Sample @0
Sample @1
Merged cross section [pb]
31.15
31.10
31.05
31.00
30.95
30.90
30.85
30.95961
31.04595
30.93824
30.97978
21.000
20.975
20.950
20.925
20.900
20.875
20.850
20.825
20.88636
20.92482
20.92978
20.88139
10.15
10.10
10.05
10.00
9.95
10.07325
10.12113
10.00846
10.09839
Old hard
Old geometric
Fresh OFF
Fresh ON
13 TeV; Qcut=70 GeV; stable W/Z; Pythia 8.317; one million events per prescription.
Point errors: conditional event/shower sampling; source integration uncertainty is separate. No cross-run covariance.
Historical and fresh shower steering differ (charged-resonance QED).

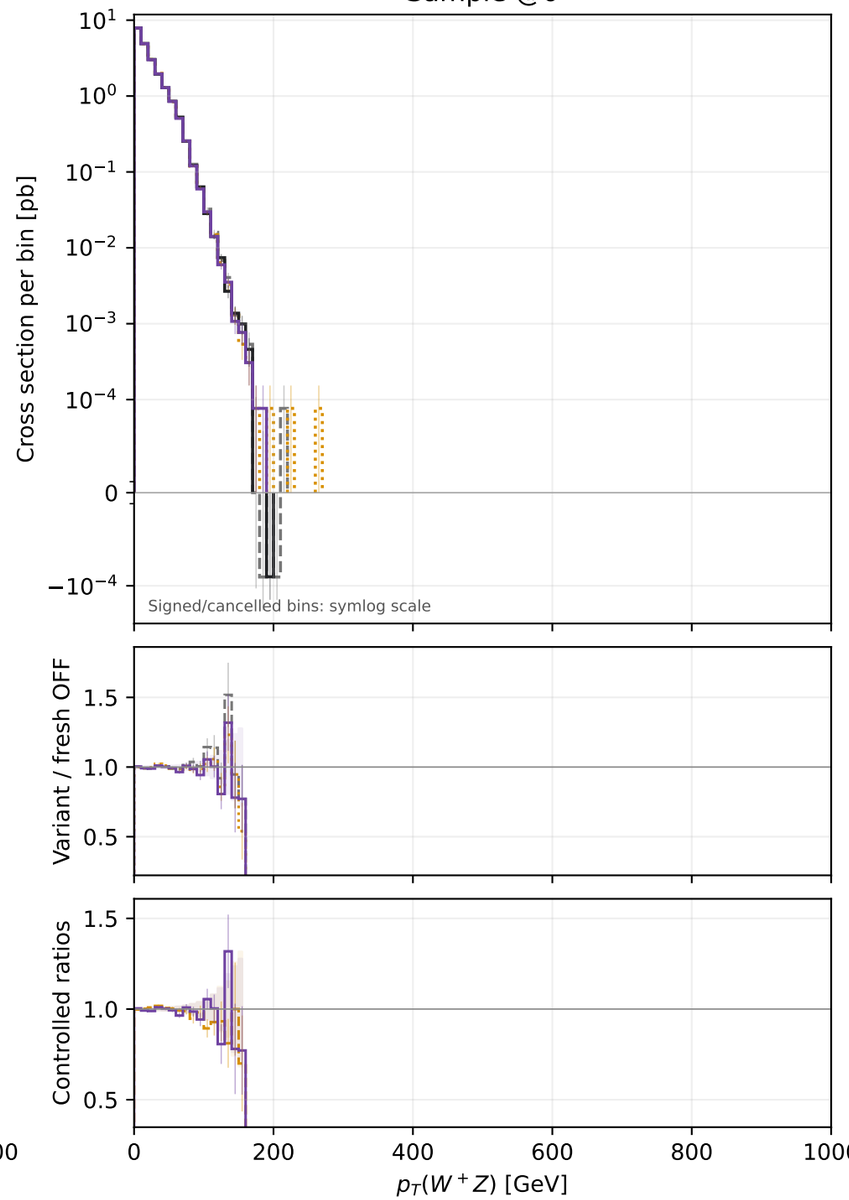
WZ + 0,1 jets: pT W+Z [GeV] — full saved range, 10 GeV bins

Original hard scale (historical)
Geometric mean (historical)
Geometric mean / ISR off (fresh)
Geometric mean / ISR on (fresh)

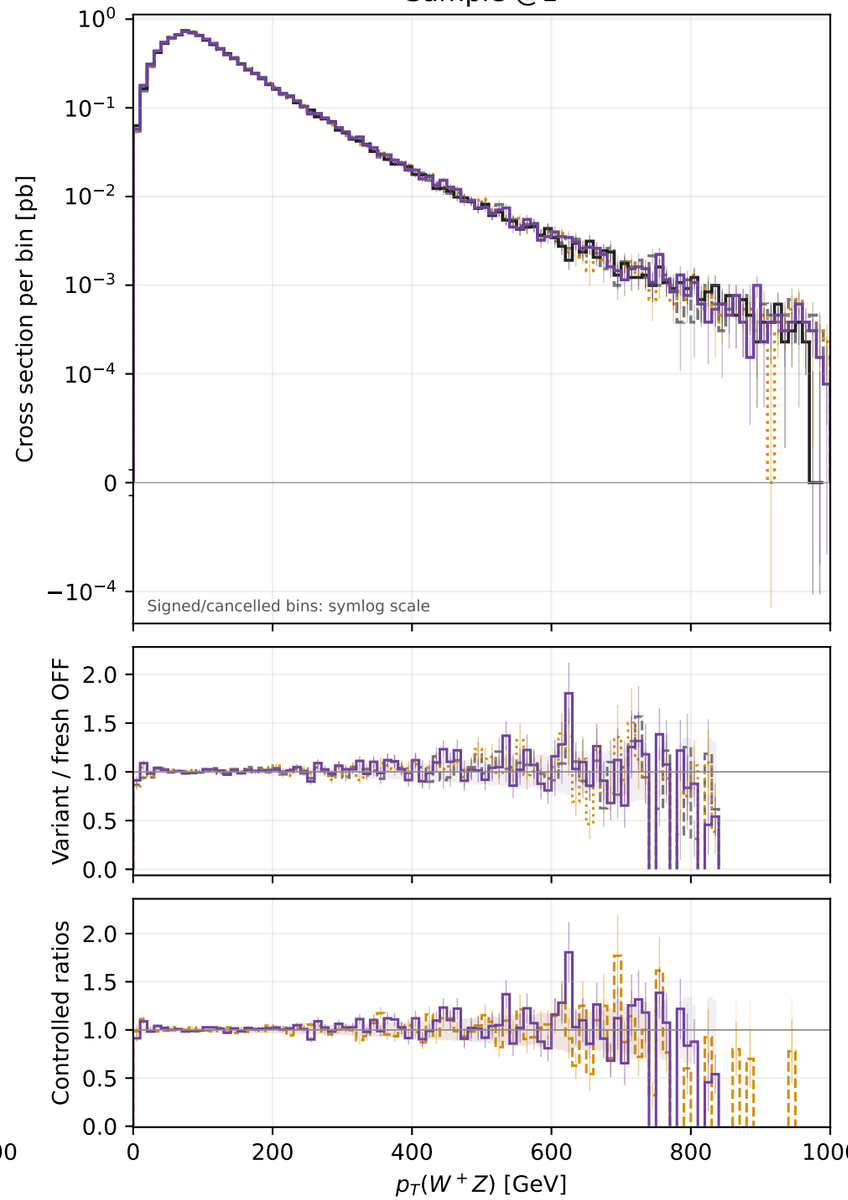
Merged



Sample @0



Sample @1



13 TeV; Qcut=70 GeV; stable W/Z; Pythia 8.317; one million events per prescription.
Bands: denominator error; bars: numerator error / denominator. Ratios require denominator > 3 statistical errors. No cross-run/bin covariance.
Errors condition on source normalization; integration-normalization errors are separate. Historical and fresh shower steering differ (charged-resonance QED).

WZ + 0,1 jets: pT W+Z [GeV] — low-scale zoom, 10 GeV bins

Original hard scale (historical)
Geometric mean (historical)
Geometric mean / ISR off (fresh)
Geometric mean / ISR on (fresh)

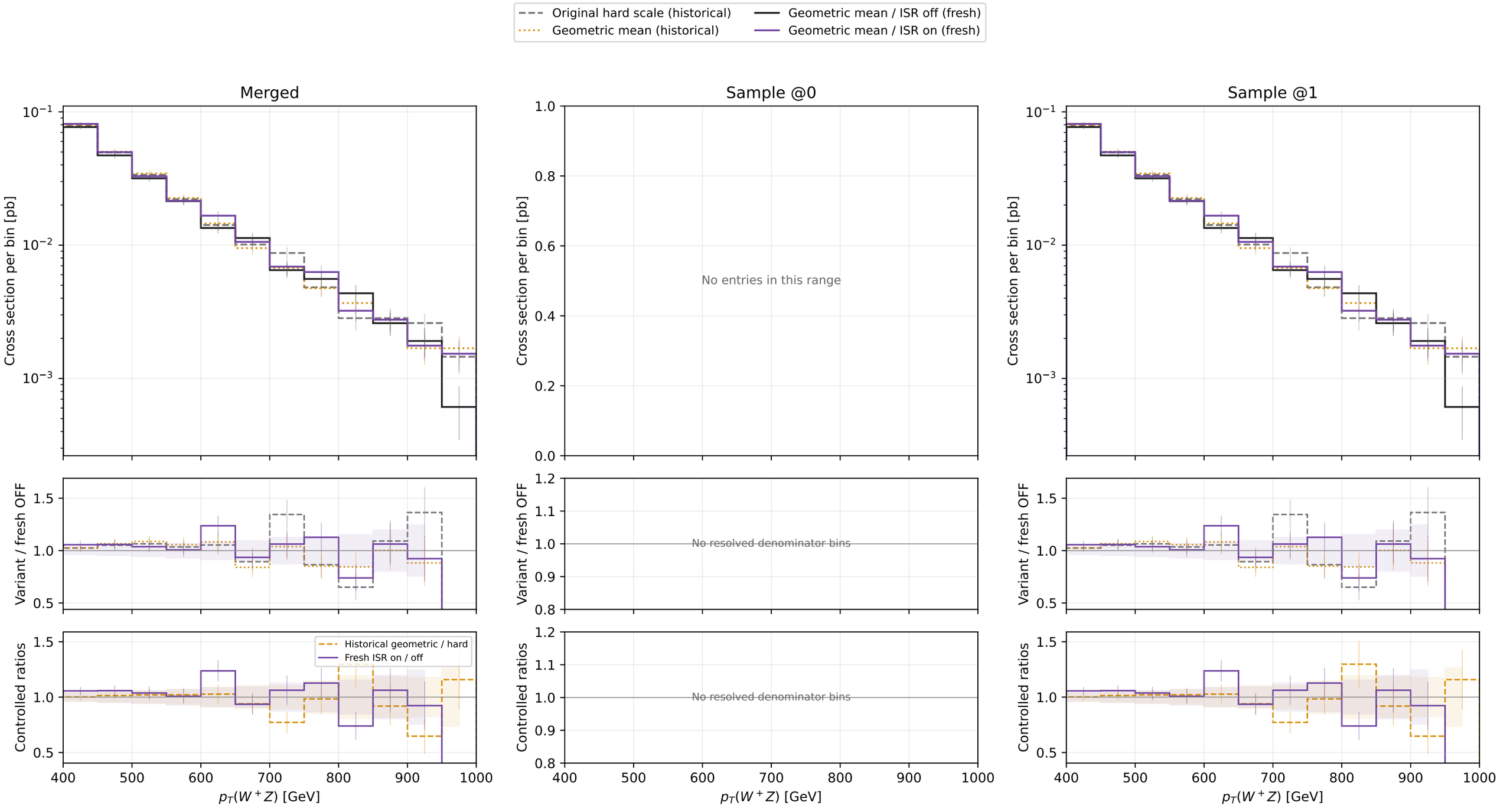
Merged
Sample @0
Sample @1

Cross section per bin [pb]
Variant / fresh OFF
Controlled ratios
Historical geometric / hard
Fresh ISR on / off

$p_T(W^+Z)$ [GeV]

13 TeV; Qcut=70 GeV; stable W/Z; Pythia 8.317; one million events per prescription.
Bands: denominator error; bars: numerator error / denominator. Ratios require denominator > 3 statistical errors. No cross-run/bin covariance.
Errors condition on source normalization; integration-normalization errors are separate. Historical and fresh shower steering differ (charged-resonance QED).

WZ + 0,1 jets: pT W+Z [GeV] — high-scale tail; rebinned, 50 GeV bins



13 TeV; Qcut=70 GeV; stable W/Z; Pythia 8.317; one million events per prescription.
Bands: denominator error; bars: numerator error / denominator. Ratios require denominator > 3 statistical errors. No cross-run/bin covariance.
Errors condition on source normalization; integration-normalization errors are separate. Historical and fresh shower steering differ (charged-resonance QED). Rebinning sums conditional bin variances; bin covariance is unavailable.

WZ + 0,1 jets: log10 pT W+Z/GeV — full saved range, 0.1-decade bins

Original hard scale (historical)
Geometric mean (historical)
Geometric mean / ISR off (fresh)
Geometric mean / ISR on (fresh)

Merged

Sample @0

Sample @1

Cross section per bin [pb]

Signed/cancelled bins: symlog scale

Variant / fresh OFF

Controlled ratios

Historical geometric / hard
Fresh ISR on / off

$\log_{10}(p_T(W^+Z)/\mathrm{GeV})$

13 TeV; Qcut=70 GeV; stable W/Z; Pythia 8.317; one million events per prescription.
Bands: denominator error; bars: numerator error / denominator. Ratios require denominator > 3 statistical errors. No cross-run/bin covariance.
Errors condition on source normalization; integration-normalization errors are separate. Historical and fresh shower steering differ (charged-resonance QED). Log histograms exclude exact zeros and underflow; they are not normalized to the total rate.

WZ + 0,1 jets: pT j1 [GeV] — full saved range, 10 GeV bins

Original hard scale (historical)
Geometric mean (historical)
Geometric mean / ISR off (fresh)
Geometric mean / ISR on (fresh)

Merged
Sample @0
Sample @1

Cross section per bin [pb]
Signed/cancelled bins: symlog scale

Variant / fresh OFF
Controlled ratios
Historical geometric / hard
Fresh ISR on / off

$p_T(j_1)$ [GeV]

13 TeV; Qcut=70 GeV; stable W/Z; Pythia 8.317; one million events per prescription.
Bands: denominator error; bars: numerator error / denominator. Ratios require denominator > 3 statistical errors. No cross-run/bin covariance.
Errors condition on source normalization; integration-normalization errors are separate. Historical and fresh shower steering differ (charged-resonance QED). Leading jets require anti-kt R=0.4, pT>20 GeV and |eta|<4.5.

WZ + 0,1 jets: pT j1 [GeV] — low-scale zoom, 10 GeV bins

Original hard scale (historical)
Geometric mean (historical)
Geometric mean / ISR off (fresh)
Geometric mean / ISR on (fresh)

Merged | Sample @0 | Sample @1

Cross section per bin [pb]

Variant / fresh OFF

Controlled ratios

Historical geometric / hard
Fresh ISR on / off

$p_T(j_1)$ [GeV]

13 TeV; Qcut=70 GeV; stable W/Z; Pythia 8.317; one million events per prescription.
Bands: denominator error; bars: numerator error / denominator. Ratios require denominator > 3 statistical errors. No cross-run/bin covariance.
Errors condition on source normalization; integration-normalization errors are separate. Historical and fresh shower steering differ (charged-resonance QED). Leading jets require anti-kt R=0.4, pT>20 GeV and |eta|<4.5.

WZ + 0,1 jets: pT j1 [GeV] — high-scale tail; rebinned, 50 GeV bins

Original hard scale (historical)
Geometric mean (historical)
Geometric mean / ISR off (fresh)
Geometric mean / ISR on (fresh)

Merged
Sample @0
Sample @1

Cross section per bin [pb]
No entries in this range
Variant / fresh OFF
No resolved denominator bins
Controlled ratios
Historical geometric / hard
Fresh ISR on / off
$p_T(j_1)$ [GeV]

13 TeV; Qcut=70 GeV; stable W/Z; Pythia 8.317; one million events per prescription.
Bands: denominator error; bars: numerator error / denominator. Ratios require denominator > 3 statistical errors. No cross-run/bin covariance.
Errors condition on source normalization; integration-normalization errors are separate. Historical and fresh shower steering differ (charged-resonance QED). Leading jets require anti-kt R=0.4, pT>20 GeV and |eta|<4.5. Rebinning sums conditional bin variances; bin covariance is unavailable.

WZ + 0,1 jets: log10 pT j1/GeV — full saved range, 0.1-decade bins

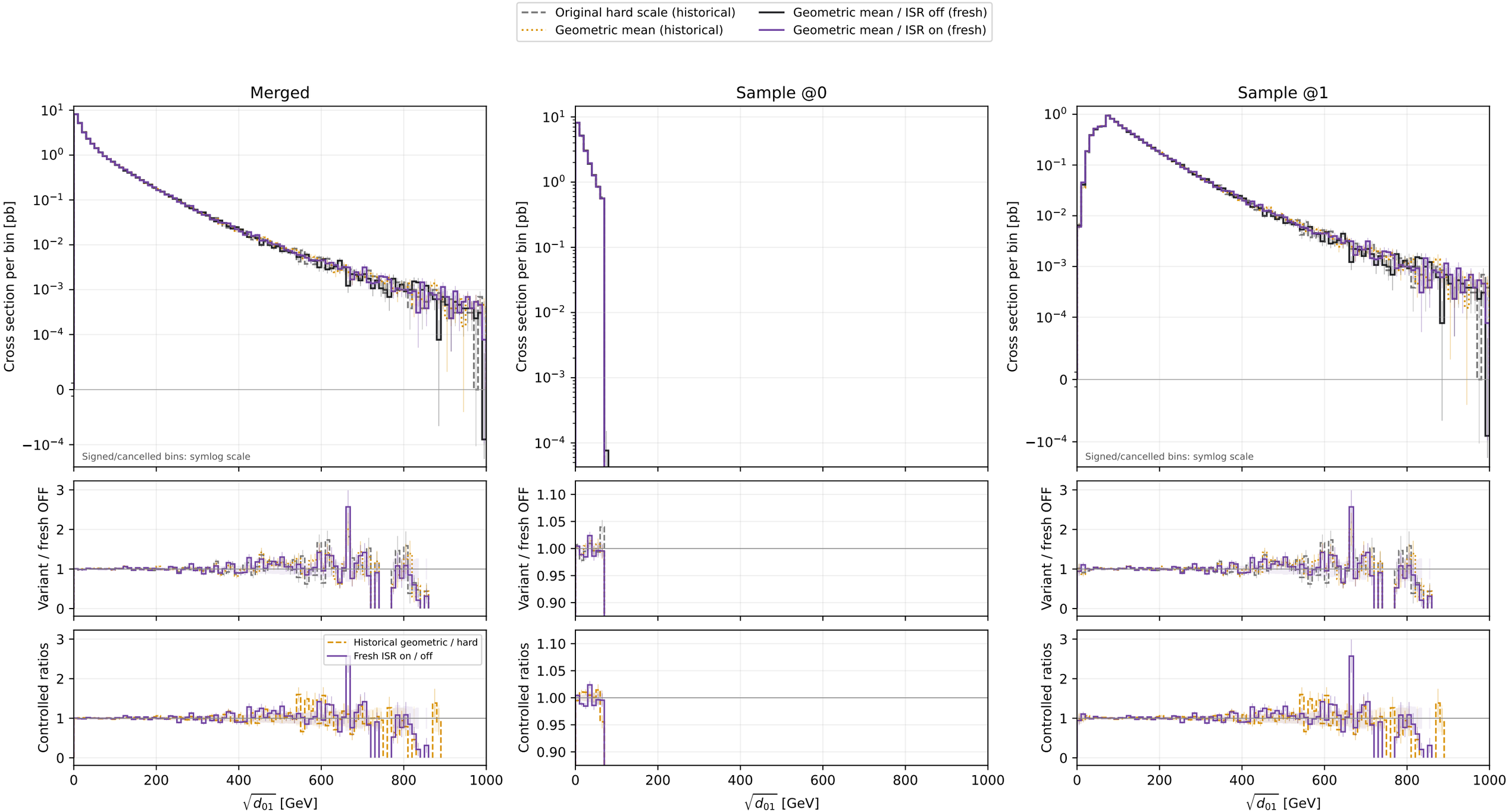
Original hard scale (historical)
Geometric mean (historical)
Geometric mean / ISR off (fresh)
Geometric mean / ISR on (fresh)

Merged
Sample @0
Sample @1

Cross section per bin [pb]
Variant / fresh OFF
Controlled ratios
Historical geometric / hard
Fresh ISR on / off
$\log_{10}(p_T(j_1)/\mathrm{GeV})$

13 TeV; Qcut=70 GeV; stable W/Z; Pythia 8.317; one million events per prescription.
Bands: denominator error; bars: numerator error / denominator. Ratios require denominator > 3 statistical errors. No cross-run/bin covariance.
Errors condition on source normalization; integration-normalization errors are separate. Historical and fresh shower steering differ (charged-resonance QED). Log histograms exclude exact zeros and underflow; they are not normalized to the total rate. Leading jets require anti-kt R=0.4, pT>20 GeV and |eta|<4.5

WZ + 0,1 jets: sqrt d01 [GeV] — full saved range, 10 GeV bins



13 TeV; Qcut=70 GeV; stable W/Z; Pythia 8.317; one million events per prescription.
Bands: denominator error; bars: numerator error / denominator. Ratios require denominator > 3 statistical errors. No cross-run/bin covariance.
Errors condition on source normalization; integration-normalization errors are separate. Historical and fresh shower steering differ (charged-resonance QED).

WZ + 0,1 jets: sqrt d01 [GeV] — low-scale zoom, 10 GeV bins

Original hard scale (historical)
Geometric mean (historical)
Geometric mean / ISR off (fresh)
Geometric mean / ISR on (fresh)

Merged
Sample @0
Sample @1

Cross section per bin [pb]
Variant / fresh OFF
Controlled ratios
Historical geometric / hard
Fresh ISR on / off
$\sqrt{d_{01}}$ [GeV]

13 TeV; Qcut=70 GeV; stable W/Z; Pythia 8.317; one million events per prescription.
Bands: denominator error; bars: numerator error / denominator. Ratios require denominator > 3 statistical errors. No cross-run/bin covariance.
Errors condition on source normalization; integration-normalization errors are separate. Historical and fresh shower steering differ (charged-resonance QED).

WZ + 0,1 jets: sqrt d01 [GeV] — high-scale tail; rebinned, 50 GeV bins

Original hard scale (historical)
Geometric mean (historical)
Geometric mean / ISR off (fresh)
Geometric mean / ISR on (fresh)

Merged
Sample @0
Sample @1

Cross section per bin [pb]
No entries in this range
Variant / fresh OFF
No resolved denominator bins
Controlled ratios
Historical geometric / hard
Fresh ISR on / off
$\sqrt{d_{01}}$ [GeV]

13 TeV; Qcut=70 GeV; stable W/Z; Pythia 8.317; one million events per prescription.
Bands: denominator error; bars: numerator error / denominator. Ratios require denominator > 3 statistical errors. No cross-run/bin covariance.
Errors condition on source normalization; integration-normalization errors are separate. Historical and fresh shower steering differ (charged-resonance QED). Rebinning sums conditional bin variances; bin covariance is unavailable.

WZ + 0,1 jets: log10 sqrt d01/GeV — full saved range, 0.1-decade bins

Original hard scale (historical)
Geometric mean (historical)
Geometric mean / ISR off (fresh)
Geometric mean / ISR on (fresh)

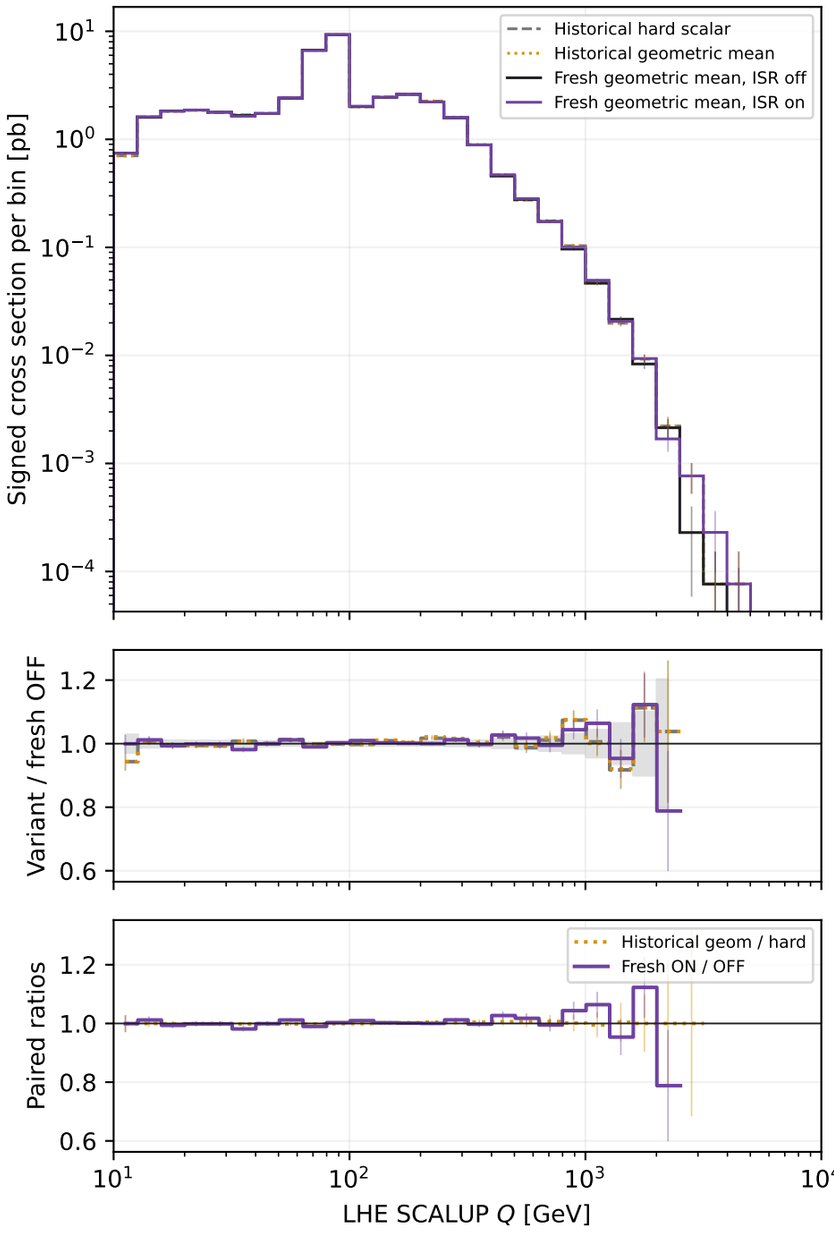
Merged
Sample @0
Sample @1

Cross section per bin [pb]
Variant / fresh OFF
Controlled ratios
Historical geometric / hard
Fresh ISR on / off
$\log_{10}(\sqrt{d_{01}}/\mathrm{GeV})$

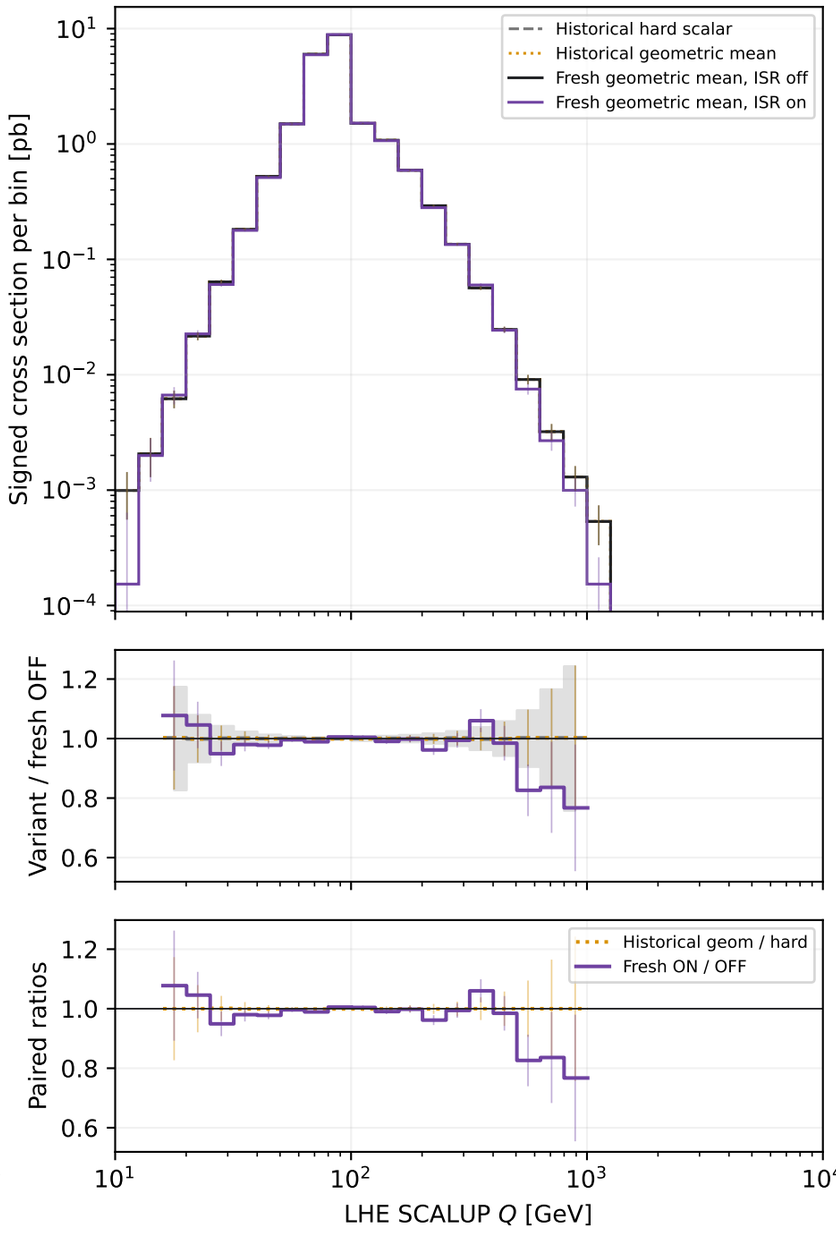
13 TeV; Qcut=70 GeV; stable W/Z; Pythia 8.317; one million events per prescription.
Bands: denominator error; bars: numerator error / denominator. Ratios require denominator > 3 statistical errors. No cross-run/bin covariance.
Errors condition on source normalization; integration-normalization errors are separate. Historical and fresh shower steering differ (charged-resonance QED). Log histograms exclude exact zeros and underflow; they are not normalized to the total rate.

WZ+0,1j source shower scales: S events before showering

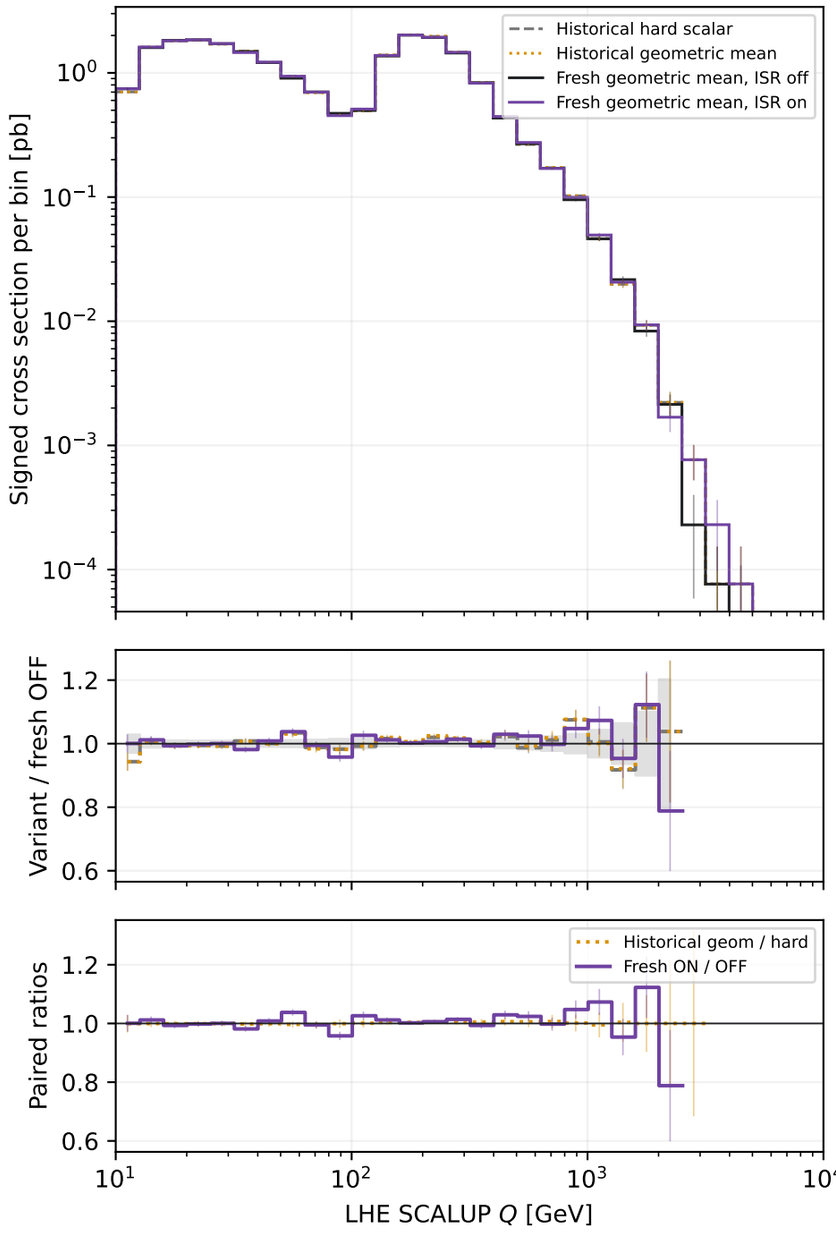
Combined source



Source @0



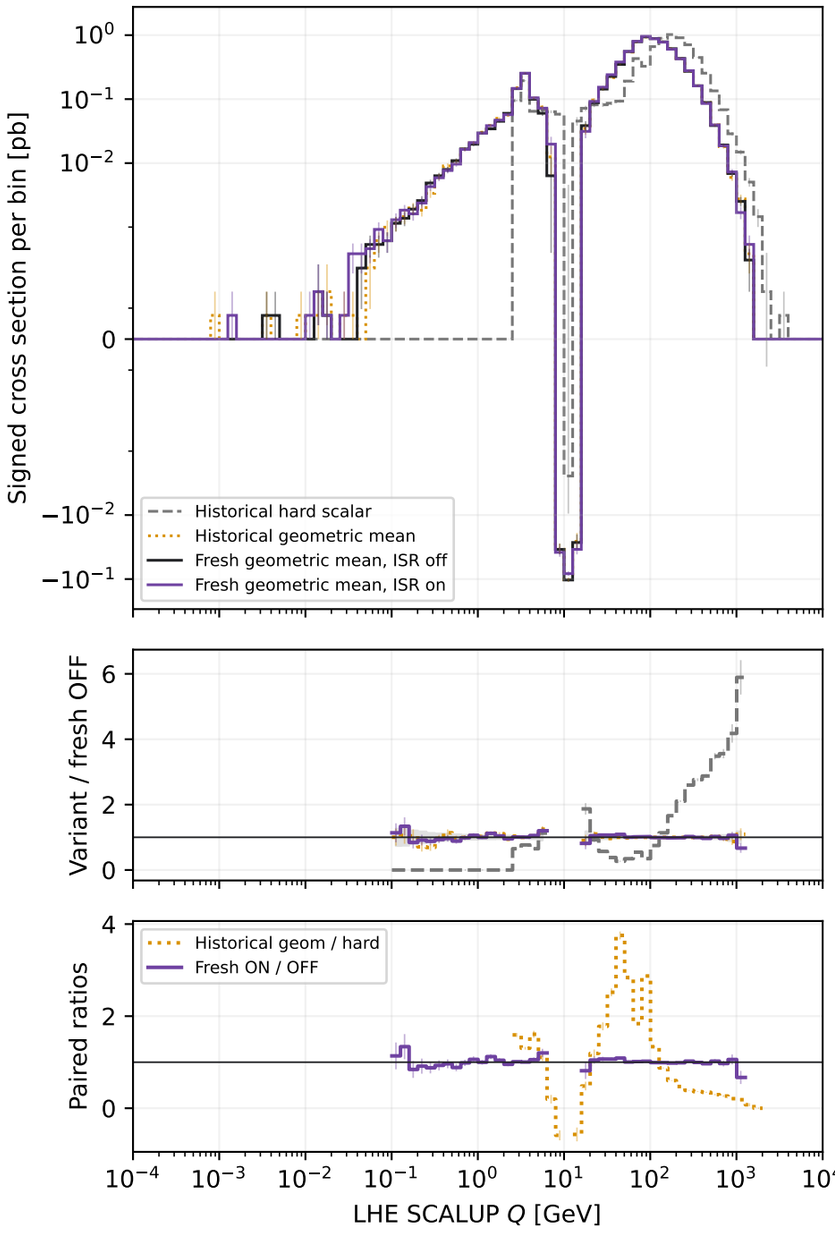
Source @1



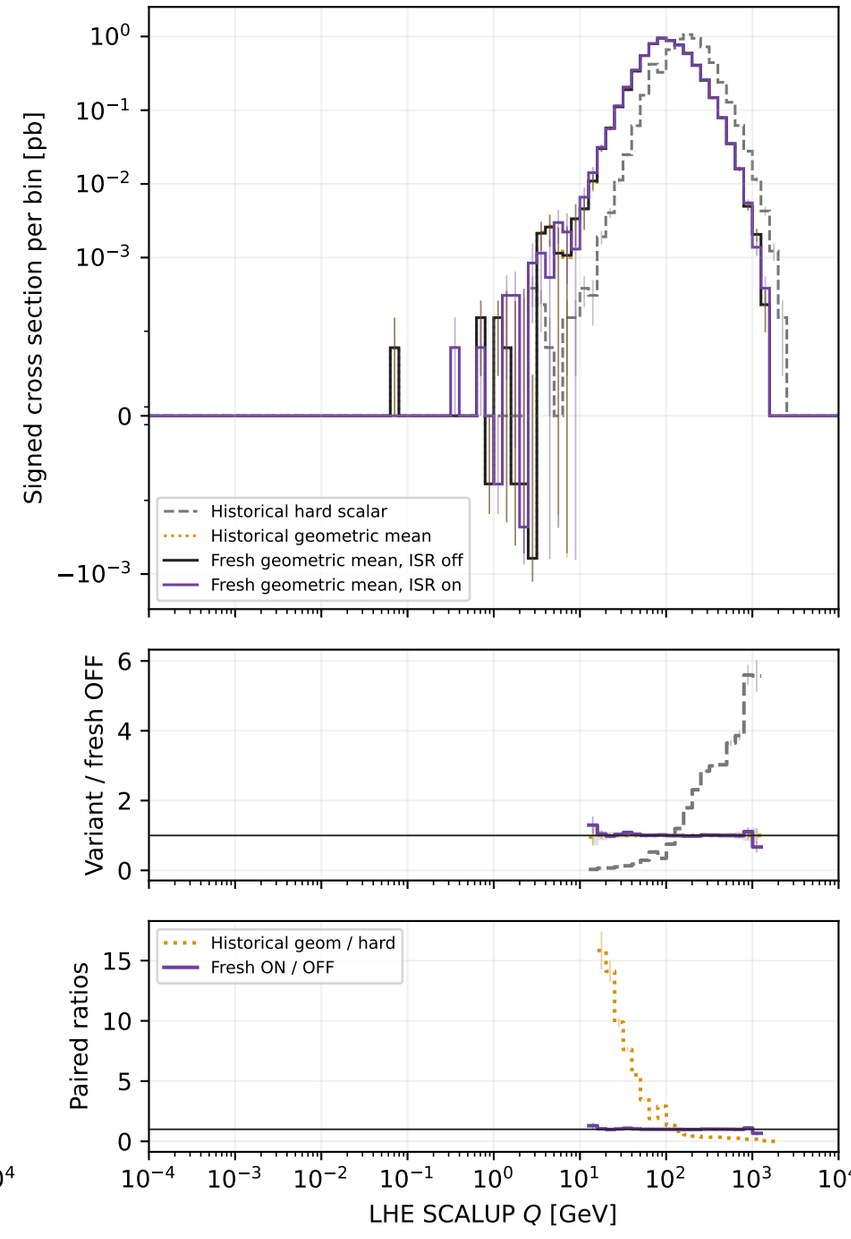
All bins retained; signed LHE weights / complete source event count; no area normalization. Ratio points require a positive denominator above 3 sampling errors.
Bars: numerator fixed-N sampling error / denominator; common-ratio band: fresh OFF. Source-normalization uncertainty and cross-run covariance are not included.
Spectrum axes use symlog where signed bins require it. Historical and fresh source epochs differ; these scales are measured before showering.

WZ+0,1j source shower scales: H events before showering

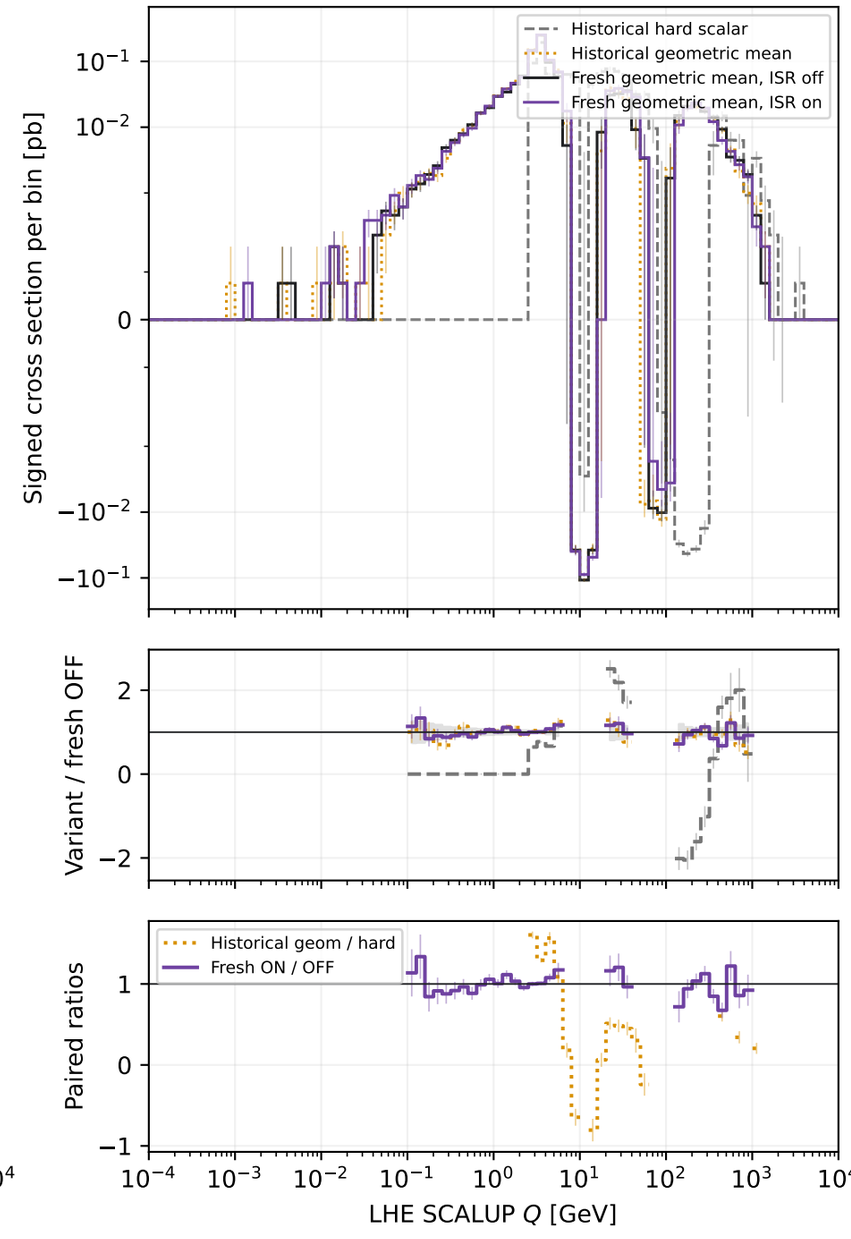
Combined source



Source @0



Source @1



All bins retained; signed LHE weights / complete source event count; no area normalization. Ratio points require a positive denominator above 3 sampling errors.
Bars: numerator fixed-N sampling error / denominator; common-ratio band: fresh OFF. Source-normalization uncertainty and cross-run covariance are not included.
Spectrum axes use symlog where signed bins require it. Historical and fresh source epochs differ; these scales are measured before showering.